

Tribhuvan University
Institute of Science and Technology
2066

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 111)
(Digital Logic)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Long Questions:

Attempt any two questions: (2 × 10=20)

1. Design the 4-bit synchronous up/down counter with timing diagram, logic diagram and truth table.
2. Design a full subtractor with truth table and logic gates.
3. Design a decimal adder with logical diagram and truth table.

Short Questions:

Attempt any eight questions: (8 × 5=40)

4. Differentiate between Analog and Digital system.
5. Convert the following octal numbers to hexadecimal.
 - a. 1760.46
 - b. 6055.263
6. Which gates can be used as inverters in addition to the NOT gate and how?
7. Draw a logic gates that implements the following
 - a) $A = (Y_1 \oplus Y_2) (Y_3 \odot Y_4) + (Y_5 \oplus Y_6 \oplus Y_7)$
 - b) $A = (X_1 \odot X_2) + (X_3 \odot X_4) + (X_4 \odot X_5) \oplus (X_4 \odot X_7)$
8. State and prove De-Morgan's theorem 1st and 2nd with logic gates and truth table.
9. Reduce the following expressions using K-map
$$\bar{A} + B(A + \bar{B} + D)(\bar{B} + C)(B + C + D)$$
10. Differentiate between a MUX and a DEMUX.
11. Explain the operation of Decoder.
12. What are the various types of shift registers?
13. What do you mean by Synchronous counter?